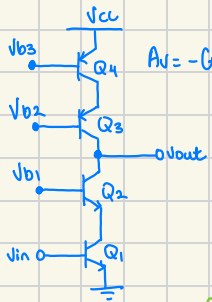
ELG 3136

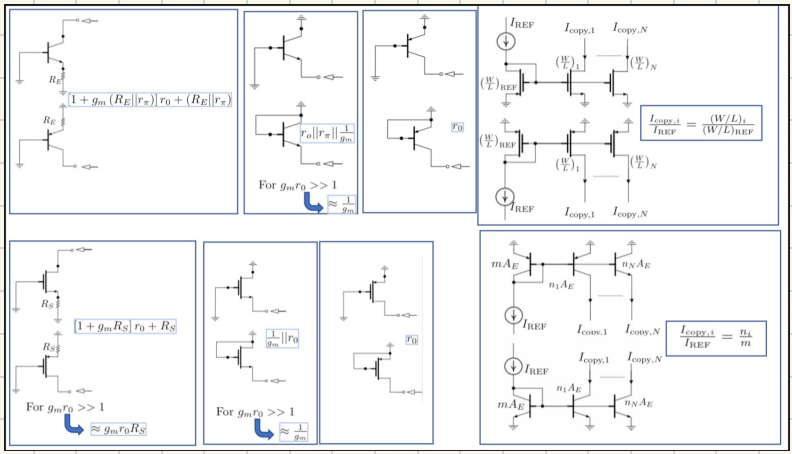
lecture 2

Finding Gain... Again

Cascode Amplifier Gain



$A_V = -G_m R_{out}$

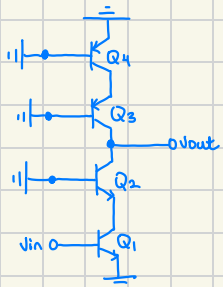


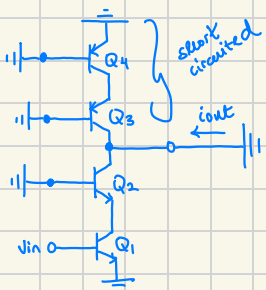
BRO CODE: $G_m = g_{m_1}$ [g_m of transistor having input voltage]

Let's prove BRO CODE and then y'all just gon have to trust me.

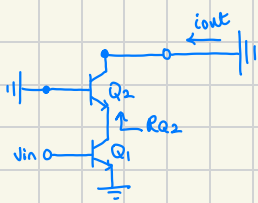
Finding G_m: ?

$$G_m = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out}=0}$$

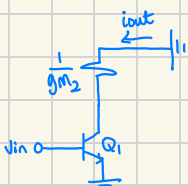




$\Rightarrow$

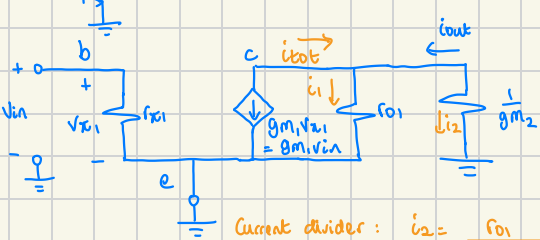


$R_{Q2} = \frac{1}{g_{m_2}}$ from circuit sheet.



Unfortunately v_{in} cannot be disabled when finding G_m since it is an AC signal.

$\Rightarrow$ Electronics 1 method.



$i_{test} = -g_{m_1}v_{in}$

$i_2 = -i_{out}$

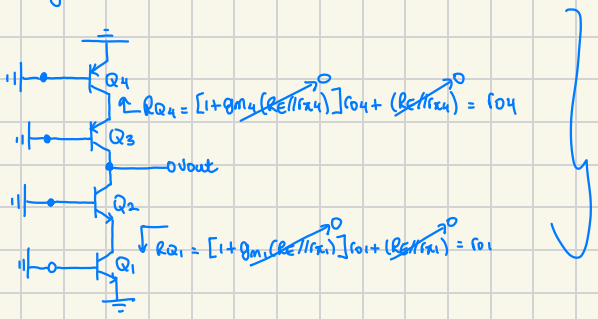
Current divider: $i_2 = \frac{r_{o_1}}{(r_{o_1} + \frac{1}{g_{m_2}})} i_{test} \Rightarrow -i_{out} = \frac{r_{o_1}}{(r_{o_1} + \frac{1}{g_{m_2}})}(-g_{m_1}v_{in})$

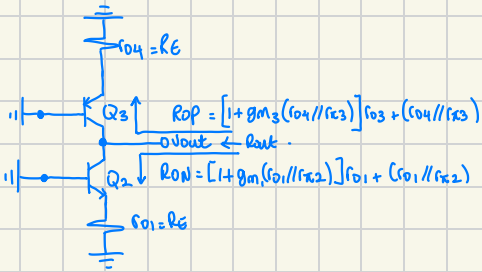
$\boxed{v_{in} = v_{x_1}}$

$\Rightarrow i_{out} = \frac{r_{o_1}}{(r_{o_1} + \frac{1}{g_{m_2}})} g_{m_1}v_{in} \Rightarrow \frac{i_{out}}{v_{in}} = G_m = g_{m_1}\left(\frac{r_{o_1}}{r_{o_1} + \frac{1}{g_{m_2}}}\right)$, Approx: $g_m r_o >> 1 \Rightarrow \boxed{r_o >> \frac{1}{g_m}}$

$\Rightarrow \boxed{G_m = g_{m_1}}$ BRO code works y'all better trust me now.

Finding Rout





$$R_{out} = R_{ON} \,//\, R_{OP} = \Big(\big[1+g_{m3}(r_{O4}\|r_{\pi3})\big]r_{O3} + (r_{O4}\|r_{\pi3})\Big) \,//\, \Big(\big[1+g_{m1}(r_{O1}\|r_{\pi2})\big]r_{O1} + (r_{O1}\|r_{\pi2})\Big)$$

$$\Rightarrow R_{out} = \Big(\big[1+g_{m3}(r_{O4}\|r_{\pi3})\big]r_{O3} + (r_{O4}\|r_{\pi3})\Big) \,//\, \Big(\big[1+g_{m1}(r_{O1}\|r_{\pi2})\big]r_{O1} + (r_{O1}\|r_{\pi2})\Big)$$

Using approx: $g_m r_O \gg 1$

$$R_{out} = \Big(\big[\cancel{1}+g_{m3}(r_{O4}\|r_{\pi3})\big]r_{O3} + (r_{O4}\|r_{\pi3})\Big) \,//\, \Big(\big[\cancel{1}+g_{m1}(r_{O1}\|r_{\pi2})\big]r_{O1} + (r_{O1}\|r_{\pi2})\Big)$$

$$= g_{m3}(r_{O4}\|r_{\pi3})r_{O3} + (r_{O4}\|r_{\pi3}) \,//\, g_{m1}(r_{O1}\|r_{\pi2})r_{O1} + (r_{O1}\|r_{\pi2})$$

$$= \Big[\big(g_{m3}r_{O3}\cancel{+1}\big)r_{O4}\|r_{\pi3}\Big] \,//\, \Big[\big(g_{m1}r_{O1}\cancel{+1}\big)r_{O1}\|r_{\pi2}\Big]$$

$$\Rightarrow \boxed{R_{out} \approx g_{m3}r_{O3}(r_{O4}\|r_{\pi3}) \,//\, g_{m1}r_{O1}(r_{O1}\|r_{\pi2})}$$

$$\Rightarrow \boxed{A_V = -G_m R_{out} = -g_{m1}g_{m3}r_{O3}(r_{O4}\|r_{\pi3}) \,//\, g_{m1}r_{O1}(r_{O1}\|r_{\pi2})}$$